

Big Mountain Report

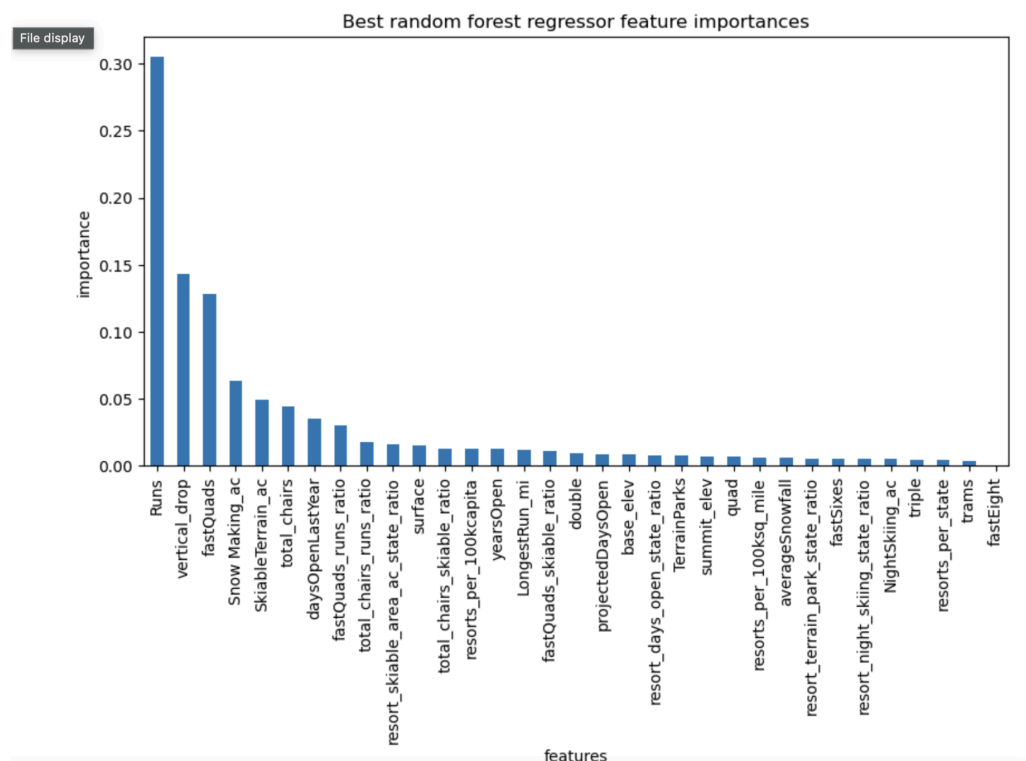
Big Mountain Ski Resort has added a new chair lift which would increase their operating cost by \$1,540,000. Currently their ticket prices are based on the market average, which has led us to look at a more data driven strategy to increase ticket pricing or cut other unnecessary expenses in order to support the expense of this new chair lift.

When wrangling through the Resort Data I noticed there was a ton of missing values. Any resort that had both weekend price and weekday price missing was dropped. Also, the fast eight feature had more than half of its values missing which I also had decided to drop since it wouldn't be as useful for evaluation. I also noticed that fast quads, fast sixes, and trams had skewed data because not all resorts had them. Wikipedia's population data was merged with the ski data in order to add state level context. The target value I ultimately decided to go forward with is the Adult Weekend price because they tend to be the same as the weekday price and it has less missing values than the Weekday price.

When further exploring the data, I realized there wasn't much categorical data so I decided to use PCA to look at some of the relationships of the features and prices. There wasn't a clear relationship between the ticket pricing and state info. After making a heat map of the correlation, it showed that the Weekend Price data is correlated with quads, chairs, vertical drop, runs, and resorts per capita. Since there wasn't a clear relationship between the state and ticket pricing I decided for modeling to include the state features integrated into the ski data(resort skiable area ac state ratio, resort night skiing state ratio, resorts per 100k capita, etc.).

After I'd explored the data, I began pre-processing and training the model. I decided to use the mean of the average ticket pricing - \$64, which gave me an r^2 of 0 and mean absolute error (MAE) of \$18-19. I used this as the baseline for any other models. I made a linear regression model that scaled data, filled the empty values with the median and selected the k best with GridSearchCV (this resulted in k=20). After plotting I'd realized using k8 will likely be the best option due to stability. The best features

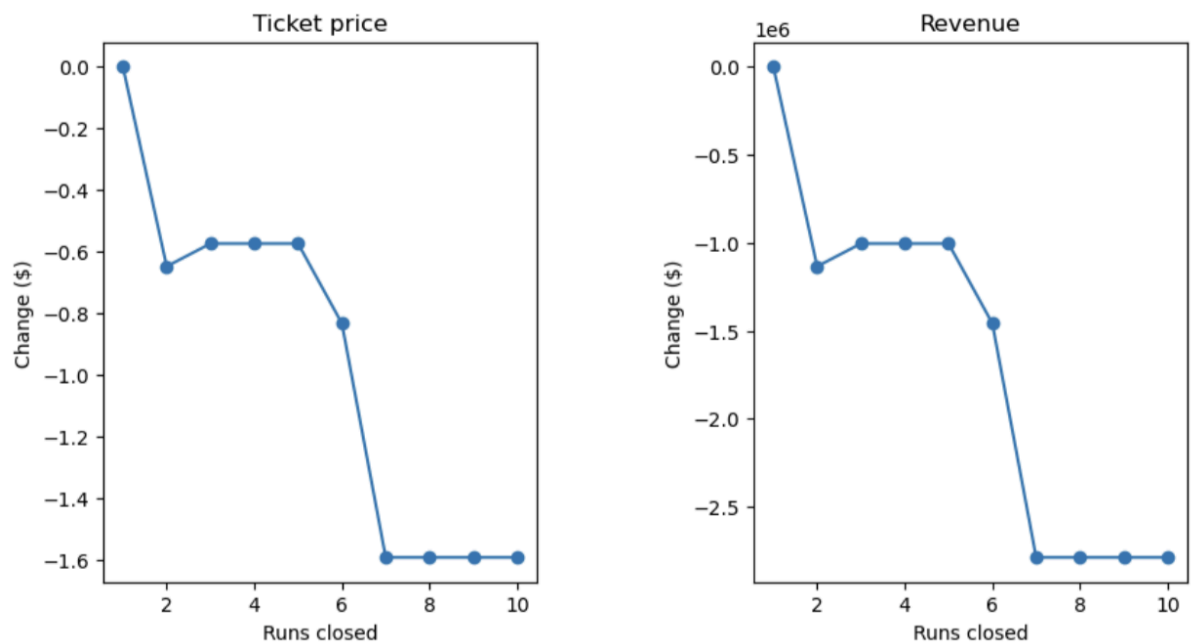
based on the coefficient were vertical drop, Fast Quads, Base Elevation, Total Chairs, and Fast Sixes. The result of the cross validation was an r^2 of 0.69 and a mean absolute error of 10.45 dollars. When run against the test, it gave an absolute error of 12.68 dollars. Although this was a big improvement from using just the mean I decided to try using a Random Forest Regressor to train the data and see if we could do better. I used Grid Search in order to get the best possible params. This led to it using 54 Trees, fill values using the mean, and scaling the data. The results of cross validation was 10.6 dollars on the training data and 9.1 dollars on the testing data. So based on it having a lower mean absolute error and smaller deviation I decided to go forward with this. The plot below shows the top features of the model are runs, vertical drop, fast quads, and snow making ac.



Big mountain Resort currently changes 81\$ and after running the model on the data, it suggest that the ticket price should be raised to \$93.04 with a mean of absolute error of \$10.41 This is a big increase, so I decided to look at how the resort compares to competitors in the important features listed in the plot above. As expected its on the higher end for all of those features but it's price was still at the market average, which would make sense as to why it was due for such an increase of \$10.41. Assuming

that each person buys 5 tickets, with the expected 350,000 visitors we would need an increase of \$0.88 in order to cover the new chairs expenses. Since it has a mean of absolute error of \$10.41 and it doesn't need much of an increase to cover the chair I'd recommend only modestly increasing ticket pricing (\$4-5) and see how consumers react.

One thing to note is that the amount of visitor per resort data was not included which could also include pricing. I also recommend for the future possibly cutting runs or excess features that don't have much an impact on pricing. For example this graph below shows how revenue would be affected based on how many rows are cut



This shows that closing 2 runs does have a small impact on price, however closing 5 still keeps revenue the same as if you closed 2. These are some things to consider for the future.